Project Overview: Parametric APDL-Based Horizontal Tailplane Structural Analysis and Optimization

This project involves structural research and design for an aircraft's **Horizontal Tailplane (HTP)**. It focuses on creating a parametric model using **ANSYS Mechanical APDL** scripts combined with aerodynamic pre-processing via **MATLAB** to analyze and optimize the structure for maximum weight reduction.

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1. Main Objectives of the Project

- **Develop Parametric APDL Scripts:** Create an HTP model that allows dimensions and structural configurations (such as wingspan, Rib/Spar/Stringer spacing, and skin thickness) to be modified by changing script variables without rebuilding the model from scratch.
- **Calculate and Distribute Loads (SFBM & BTL):** Use MATLAB to calculate shear forces and bending moments (Shear Force & Bending Moment), as well as Balancing Tail Load (BTL) in compliance with aviation safety standards **EASA CS-25** (referenced from the ATR 42-600 aircraft).
- **Weight Reduction and Optimization:** Compare multiple structural configurations to minimize the total mass/weight of the HTP as much as possible, while maintaining structural integrity and safety margins (Factor of Safety > 1.5).

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2. Current Project Status (Schedule & Milestone)

Based on the progress tracking sheet (*Project Schedule Tracking*):

- **Overall Task Status:** A total of **10 Tasks** (3 completed, 1 in progress, and 6 pending).
- **Current Status (as of 29 Sep 2026):**
  - **Completed (T-01 to T-03):** Topic study/literature review, data collection, and parameter definition (Stage 1)
  - **In Progress (T-04):** Parametric APDL script development (Stage 2a) from 28–29 Sep 2026
  - **Next Milestone (T-05):** Proposal presentation for project approval on **30 Sep 2026**

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3. Structure and Methodology

1. **Stage 1: Input Data Definition**
   - Utilizes geometry and flight envelope data from the **ATR 42-600** (e.g., MTOW 18,600 kg, V_C/V_MO speed 250 KIAS) to synthesize aerodynamic loads.
   - Applies material properties of **Aluminum 7075-T6** for internal structures (Ribs, Spars, Stringers) and equivalent isotropic properties for composite skins.
2. **Stage 2: APDL Scripting & Load Reconstruction**
   - Develops APDL scripts to generate geometry using *DO loops and mesh using SHELL181 and BEAM188 elements.
   - Uses MATLAB to solve the Least-Norm problem to map bending and shear loads

onto FEA nodes.

3. **Stage 3: Configurations & Verification**
    - **Configuration A (Baseline):** 14 Ribs, 3 Spars, 8 Stringers (referenced from the HERFUSE study)
    - **Configuration B & C:** Reduces the number of Ribs and Spars to find the lightest layout that still satisfies strength criteria.
    - **Model Verification:** Validates the model using Free-Free Modal Analysis (requiring 6 rigid-body modes) and 1g gravity load testing.

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4. Budget

- **Net Budget: 0.00 THB (Fully Funded / Zero Cost)**
- All software and hardware utilize university licenses and educational versions (ANSYS Student Version, MATLAB University License, Personal Workstations).

Sources:

- [Proposal] Parametric APDL-Based Horizontal Tailplane Structural Analysis and Optimization.pdf
- Project Schedule Tracking
- Parameter Summarize ATR 42-400-(600)
- Finished Stage 1: Data Collection & Parameter Definition
- [Refence Literature] A Parametric APDL-Based Workflow for Efficient Structural Design of a Civil Aircraft Tailplane.pdf